UNIVERSITY OF ECONOMICS AND BUSINESS – VNU

Faculty of Accounting and Auditing

FINALASSIGNMENT

COURSE: DATA ANALYSIS IN ACCOUNTING

TOPIC: STOCK PRICES OF MASTERCARD AND VISA (2008 – 2004) ANALYSIS

Instructor	Dr. Vũ Thị Thanh Bình
Student	Nguyễn Thị Hải Vân
Date of Birth	22/09/2004
Student ID	22050912
Class	QH-2022-E KTKT 4
Major	Business Analytics
Course Code	FAA4023-E

HA NOI – JUNE 2025

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LIST OF ABBREVIATIONS

Abbreviations	Meaning
AI	Artifical Information
Adj	Adjusted
V	Visa Card
M	MasterCard
RQ	Research Question
RO	Research Objective
U.S.	United States

CHAPTER 1. OVERVIEW OF THE RESEARCH

I. Brief description of the company/dataset.

1.1. Description of the company

- Visa and MasterCard are among the most popular international payment cards worldwide, owned by Visa International Service Association and MasterCard Worldwide, both headquartered in the United States. However, Visa and MasterCard do not issue credit cards directly. Instead, they provide payment systems to banks and financial institutions, partnering with them to issue Visa and MasterCard cards to their customers.

1.2. Description of the dataset

- This dataset contains historical stock price data for MasterCard (MA) and Visa (V) from June 2008 to June 2024. The data includes daily stock prices with the following columns:

- Date: The date of the recorded stock prices.
- Open_M: Opening price of MasterCard stock.
- High_M: Highest price of MasterCard stock for the day.
- Low_M: Lowest price of MasterCard stock for the day.
- Close_M: Closing price of MasterCard stock.
- Adj Close_M: Adjusted closing price of MasterCard stock.
- Volume_M: Trading volume of MasterCard stock.
- Open_V: Opening price of Visa stock.
- High_V: Highest price of Visa stock for the day.
- Low_y: Lowest price of Visa stock for the day.
- Close_y: Closing price of Visa stock.
- Adj Close_y: Adjusted closing price of Visa stock.
- Volume_y: Trading volume of Visa stock.

II. Research Question and Research Objectives

2.1. Research Question

- RQ1. How has the stock price of Visa and MasterCard changed over time? Are there any notable differences in their price trends?

- RQ2. How has the trading volume of Visa and MasterCard stocks fluctuated over time?
- RQ3. How have the highest and lowest prices per day of Visa and MasterCard changed during different economic periods?
- RQ4. What is the relationship between trading volume and the highest and lowest daily stock prices of Visa and MasterCard? Does the change in trading volume affect the fluctuations in daily stock prices?
- RQ5. Based on historical stock price data, what are the expected trends in the stock prices of Visa and MasterCard from July 2024 to June 2026? How do these trends compare to the historical data?
- RQ6. What would be the recommended investment action (Buy, Hold, Sell) for Visa and MasterCard stocks?
- RQ7. How does the historical trading volume impact the fluctuations in stock prices of Visa and MasterCard, and what role does trading volume play in the decision to Buy, Hold, or Sell?
- RQ8. What are the potential risks and opportunities identified from the historical data of Visa and MasterCard, and how can investors leverage these insights to make informed investment decisions going forward?

2.2. Research Objective

- RQ1. How has the stock price of Visa and MasterCard changed over time? Are there any notable differences in their price trends?

- RO1. Analyze the stock price volatility of Visa and MasterCard over the years, using price indicators such as opening price, closing price, highest price, and lowest price. Identify the main patterns and trends in stock price fluctuations and evaluate the differences in trends between the two companies.

- RQ2. How has the trading volume of Visa and MasterCard stocks fluctuated over time?

- RO2. Analyze the changes in the trading volume of Visa and MasterCard stocks over the years. Assess the impact of trading volume on stock prices and identify fluctuations in trading volume during different periods.

- RQ3. How have the highest and lowest prices per day of Visa and MasterCard changed during different economic periods?

- RO3. Study the fluctuations in the highest and lowest trading prices of Visa and MasterCard stocks per day. Identify periods with significant price variations and analyze the factors influencing these fluctuations.
- RQ4. What is the relationship between trading volume and the highest and lowest daily stock prices of Visa and MasterCard? Does the change in trading volume affect the fluctuations in daily stock prices?
- RO4. Analyze the relationship between trading volume and the highest and lowest daily stock prices of Visa and MasterCard. Investigate whether changes in trading volume have an impact on the stock price volatility.
- RQ5. Based on historical stock price data, what are the expected trends in the stock prices of Visa and MasterCard from July 2024 to June 2026? How do these trends compare to the historical data?
- RO5. For this data I will use Python to analyze the closing price and volume of Visa and MasterCard and compare it with available data
- RQ6. . What would be the recommended investment action (Buy, Hold, Sell) for Visa and MasterCard stocks?
- RO6. Analyze whether holding or selling stocks with such indicators would be a reasonable decision for investors.
- RQ7. How does the historical trading volume impact the fluctuations in stock prices of Visa and MasterCard, and what role does trading volume play in the decision to Buy, Hold, or Sell?
- RO7. Analyze the impact of historical trading volume on stock price fluctuations for Visa and MasterCard and how it influences investment decisions (Buy, Hold, Sell) such as examine how trading volume correlates with stock price changes for both companies, investigate how trading volume reflects market sentiment and investor behavior and analyze how trading volume affects decisions to Buy, Hold, or Sell
- RQ8. What are the potential risks and opportunities identified from the historical data of Visa and MasterCard, and how can investors leverage these insights to make informed investment decisions going forward?
- RO8. To identify potential risks and opportunities from the historical data of Visa and MasterCard and guide investors in making informed decisions.

CHAPTER 2. METHODOLOGY

I. Data collecttion

- I collected this dataset from Kaggle, a popular platform for data science competitions and datasets. Kaggle offers a wide variety of publicly available datasets for research and analysis. This particular dataset includes historical stock prices for Visa and MasterCard from June 2008 to June 2024. It consists of key information such as daily stock prices, trading volumes, and price data for both companies. The data includes daily stock prices with the following columns:

- Date: The date of the recorded stock prices.
- Open_M: Opening price of MasterCard stock.
- High_M: Highest price of MasterCard stock for the day.
- Low_M: Lowest price of MasterCard stock for the day.
- Close_M: Closing price of MasterCard stock.
- Adj Close_M: Adjusted closing price of MasterCard stock.
- Volume_M: Trading volume of MasterCard stock.
- Open_V: Opening price of Visa stock.
- High_V: Highest price of Visa stock for the day.
- Low_y: Lowest price of Visa stock for the day.
- Close_y: Closing price of Visa stock.
- Adj Close_y: Adjusted closing price of Visa stock.
- Volume_y: Trading volume of Visa stock.

- The dataset allows for a detailed analysis of the historical performance of Visa and MasterCard stocks, enabling the identification of trends, fluctuations, and insights into how these companies' stock prices have behaved over time.

II. Master the data

2.1. Convert data

- The initial Date format of this dataset was in Whole Number format, so I right-clicked on the Date column and selected the Date option to change the format to DD/MM/YY.

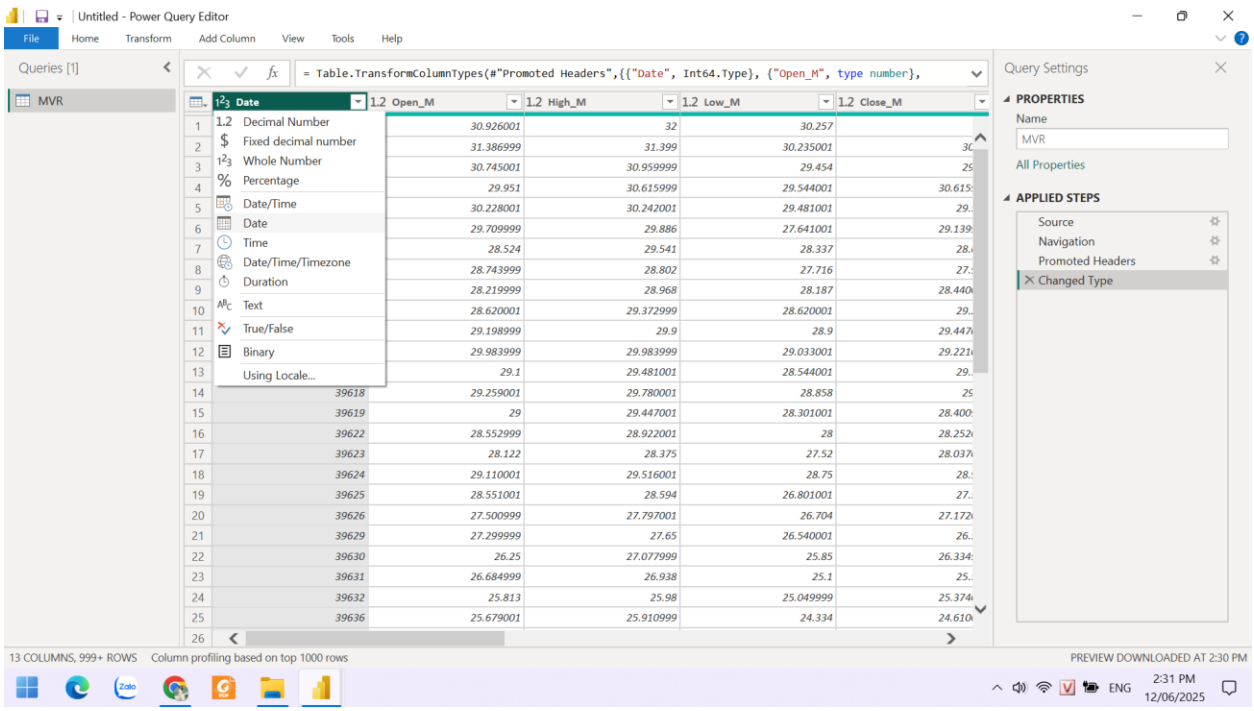


Figure 2.1. Right-clicked on the Date column

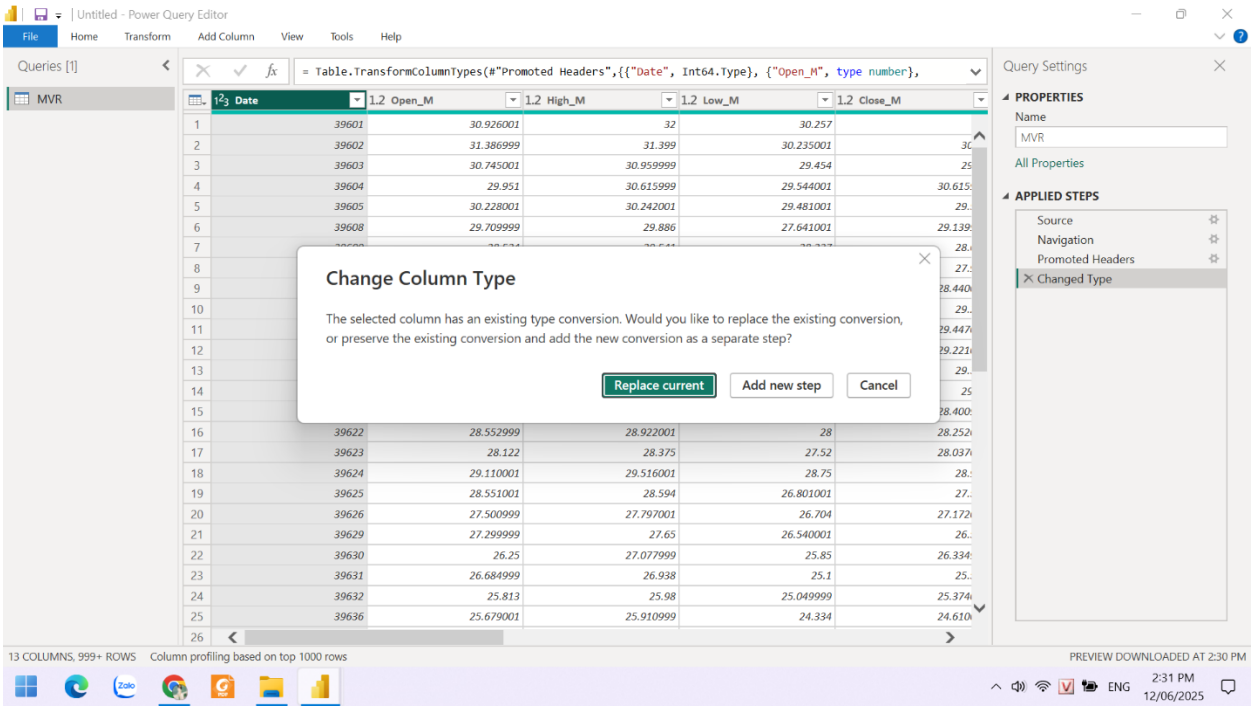


Figure 2.2. Replace Date

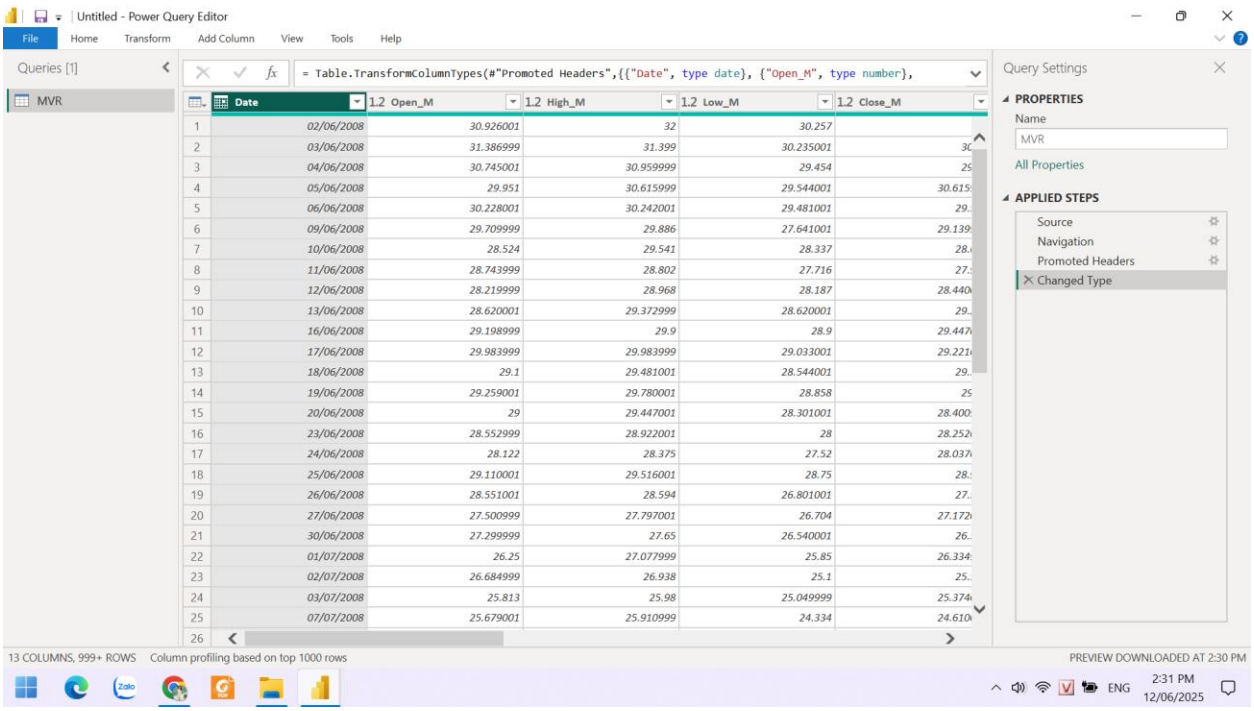


Figure 2.3. Column Date

2.2. Data Segmentation

- In this step, I clicked on the Date column, went to the Add Column tab, clicked on the arrow next to Date in the From Date & Time section, and selected Year. Then, I did the same for Month and Day.

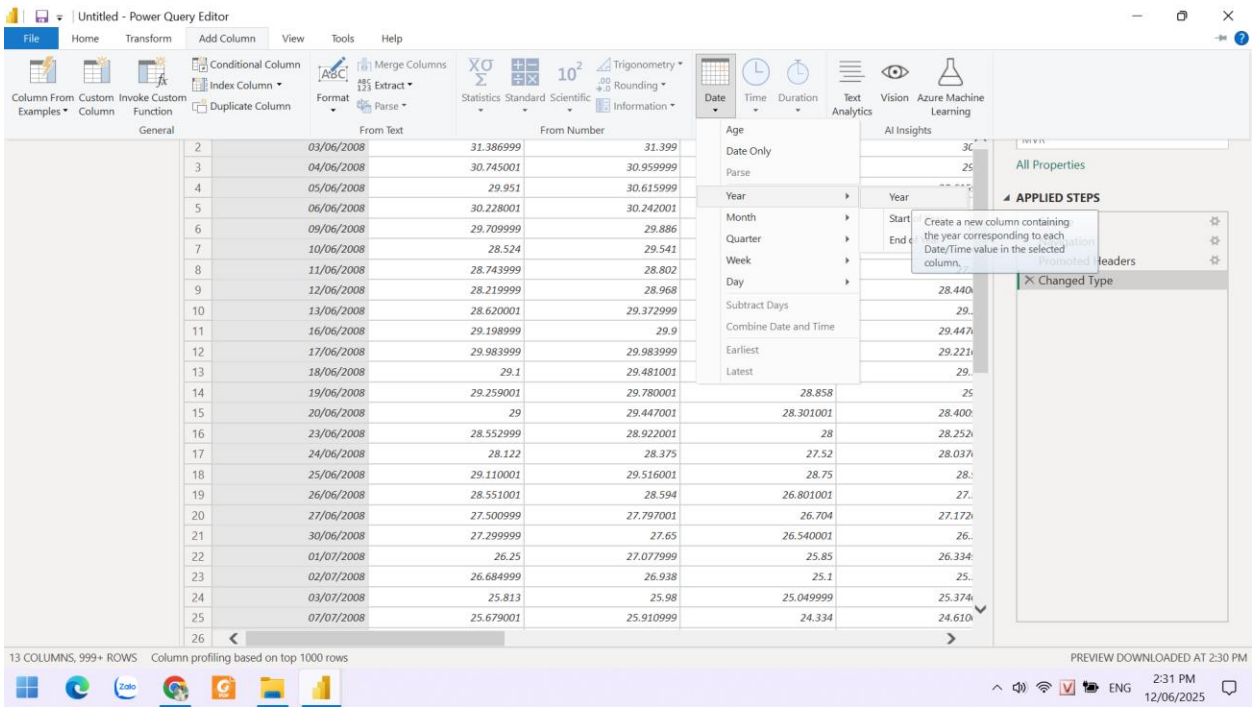


Figure 2.4. Split column Year

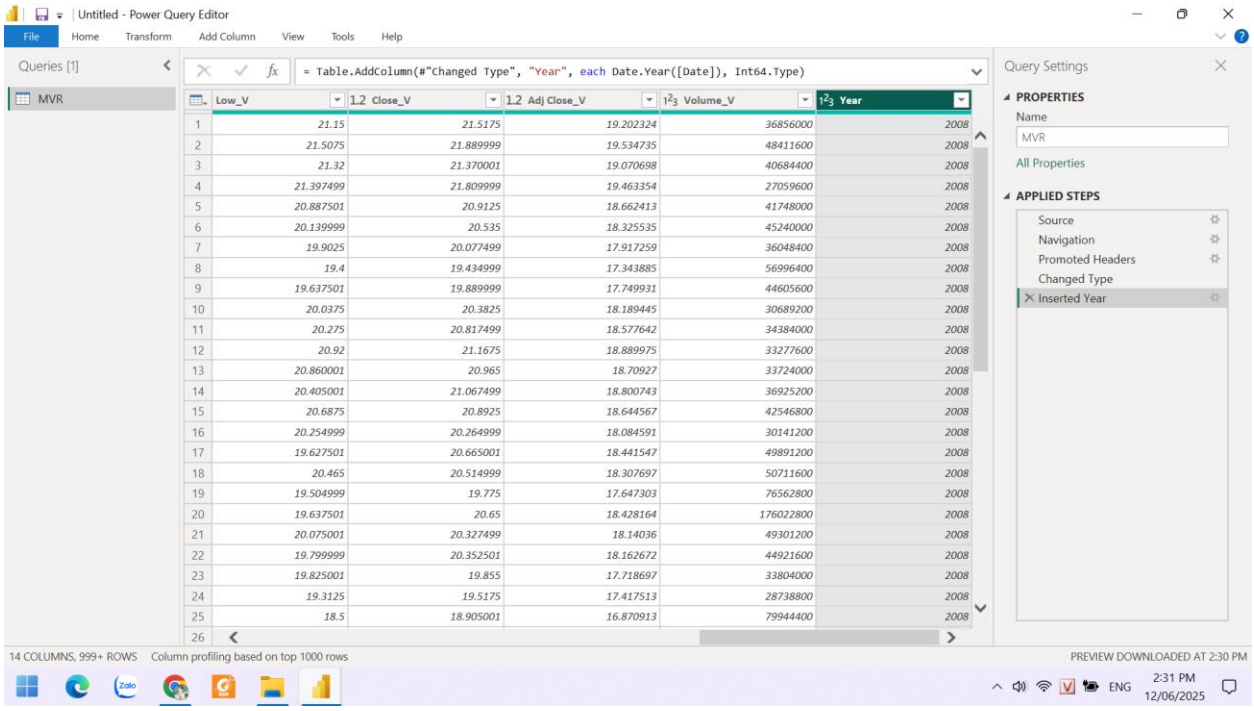


Figure 2.5. Column year

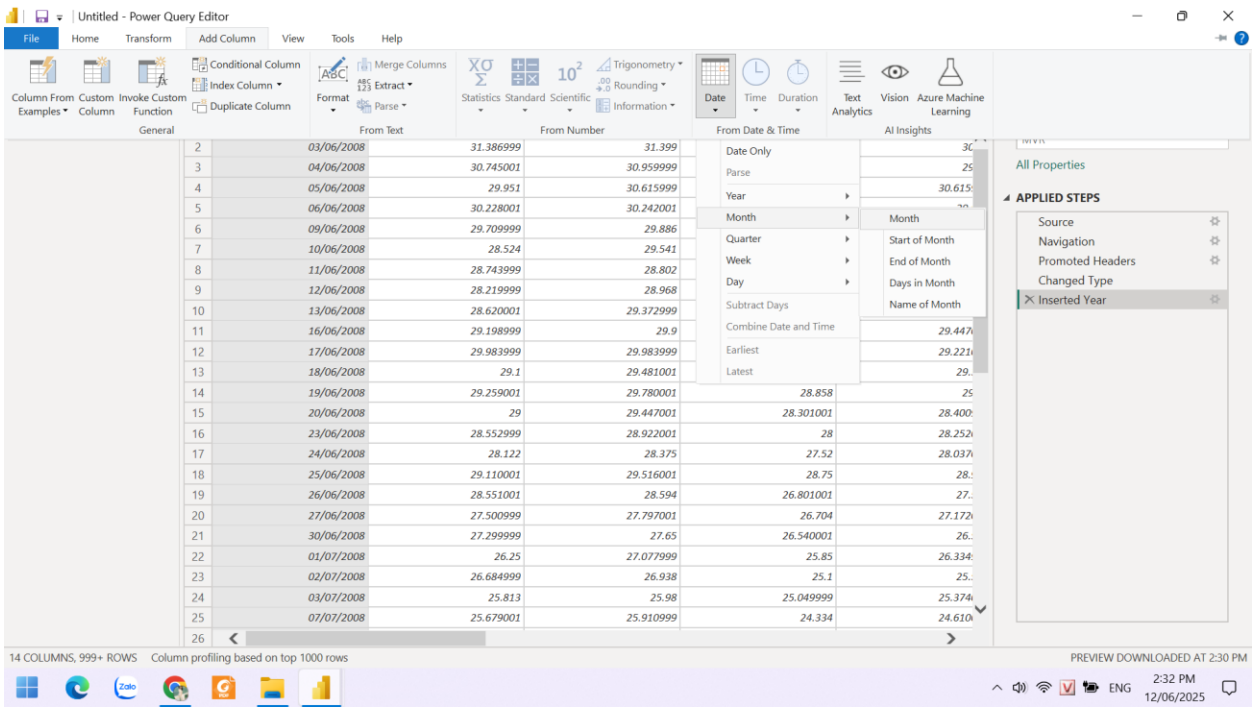


Figure 2.6. Split column Month

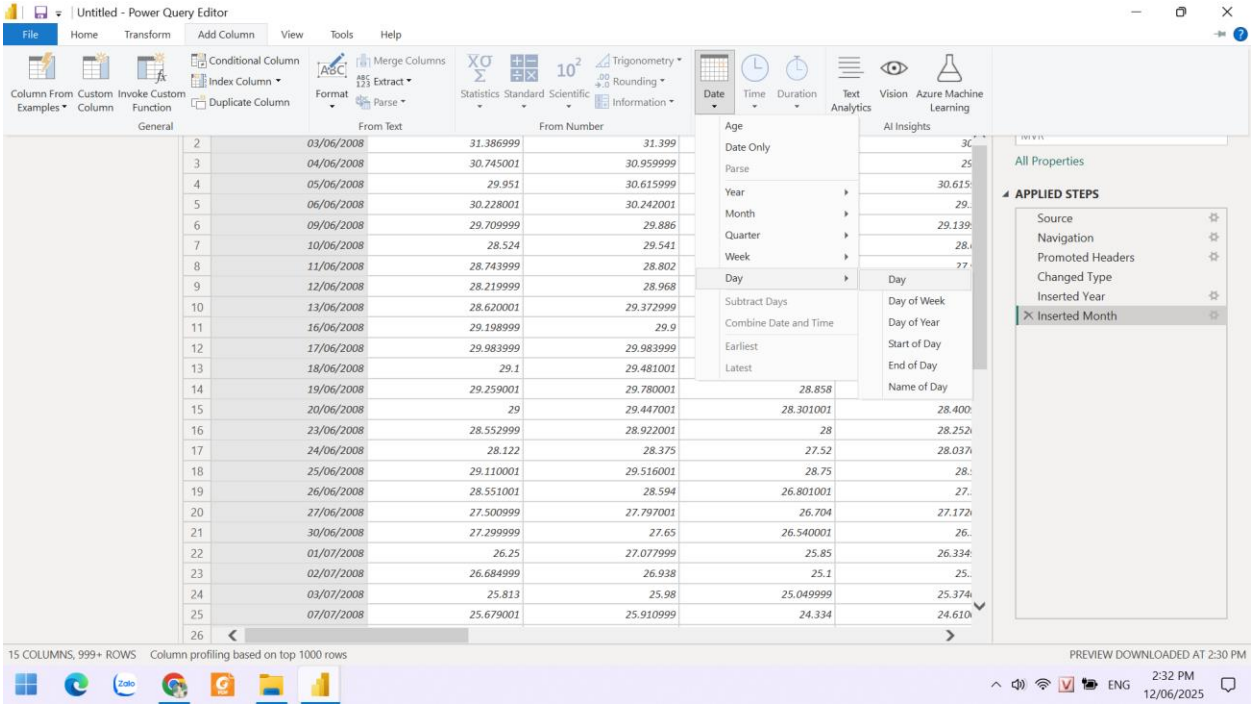


Figure 2.7. Split column Day

III. Data information

3.1.Type of data

- Time Series Data: The data primarily includes the stock prices of Visa and MasterCard over time, meaning it is formatted by Date, corresponding to specific time points (by year, month, and day).
- Numeric Data: Indicators such as the opening price, closing price, highest price, lowest price, and trading volume are numeric values. In this dataset, values like the opening price, closing price, highest price, and lowest price are formatted as floating-point numbers, while the trading volume is formatted as integer data.

3.2.Type of business analytics approaches

- For this project, I applied Power BI as the tool for analysis and combined two methods: Descriptive Analytics and Predictive Analytics.

3.2.1. Descriptive Analytics

- For this method, I used Power BI to create various visualizations, such as column charts, line charts, and stacked area charts, to illustrate the trends, changes over time, and key characteristics of stock prices. These visualizations help to identify and better understand the long-term movements and fluctuations in the stock prices of Visa and MasterCard.
- Additionally, I created cards to display important metrics such as Open, Close, and Volume, which allows for easy reference to key data points and provides a clear summary of the stock's performance over the selected period.
- A notable feature is that I linked these cards to the visualizations below (using slicers to analyze by year, month, and day). When adjusting the slicers below the charts, these cards also dynamically update accordingly.

3.2.2. Predictive Analytics

- For this analysis, I utilized Python to forecast the stock price and trading volume of Visa and MasterCard up until June 2026.
- The reason I chose the closing price is that it is a key indicator of stock performance and widely used in financial analysis.

- Additionally, I incorporated the trading volume data because it provides insight into market activity and investor sentiment, which can influence stock price movements. The forecast is based on historical data and helps to predict future trends, allowing investors to make more informed decisions.
- The model used (ARIMA) captures the underlying patterns in both stock price and trading volume, accounting for past trends and market behavior.
- Additionally, I will use Diagnostic Analytics and Prescriptive Analytics, but these two methods are not the primary focus of my analysis.

3.2.3. Diagnostic analytics

- In this dashboard, I analyzed the differences between high, low and volume, and I will also analyze the relation between volume and stock price fluctuations on the charts I created. This will help identify whether there are any differences, and if so, what the causes might be.

3.2.4. Prescriptive Analytics

- For this method, based on the existing data, I will provide actionable recommendations regarding whether investors should hold or sell their Visa and MasterCard stocks.
- This is done by analyzing the forecasted stock prices, trading volumes, and the general market trends.
- These insights can guide investors on how to respond to future price movements and trading conditions, considering factors such as expected price growth, volatility, and volume trends.
- The goal is to assist investors in optimizing their investment strategies to maximize returns while minimizing risk.

CHAPTER 3. THE RESEARCH FINDINGS

I. Stock Price Trends of Visa and MasterCard

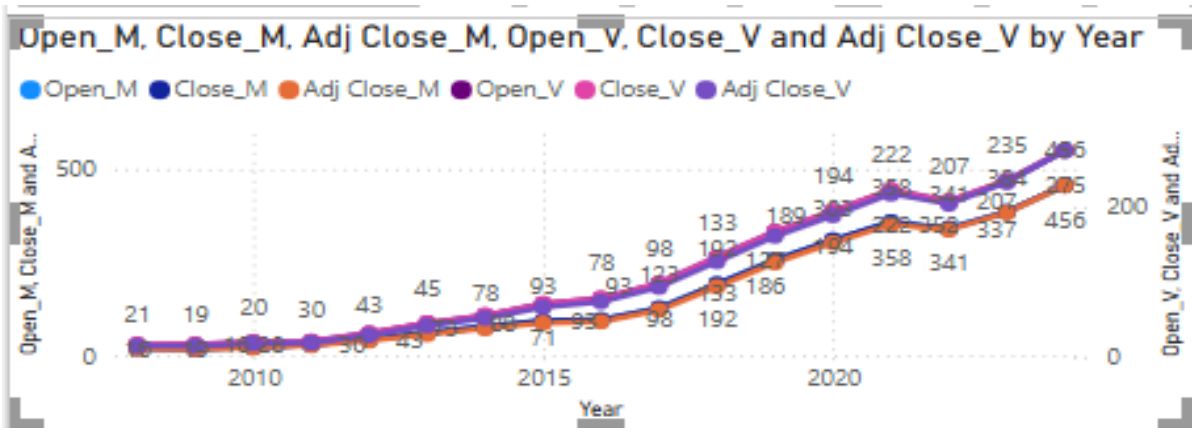


Chart 3.1. Stock Price Trends of Visa and MasterCard

- Visa Stock Trends: The historical data for Visa shows a steady and strong increase in stock price over the years, particularly after the 2010s. The stock price demonstrates
- significant growth with periodic fluctuations, which are consistent with market trends. The forecasting data suggests that Visa's stock price is expected to continue increasing, reaching a steady level by mid-2024, with minimal growth thereafter.
- A noticeable drop can be seen in the first half of quarter 2 2024. The cause could be due to market conditions and the changes in consumer behavior, particularly in the lower-spending consumer segment.

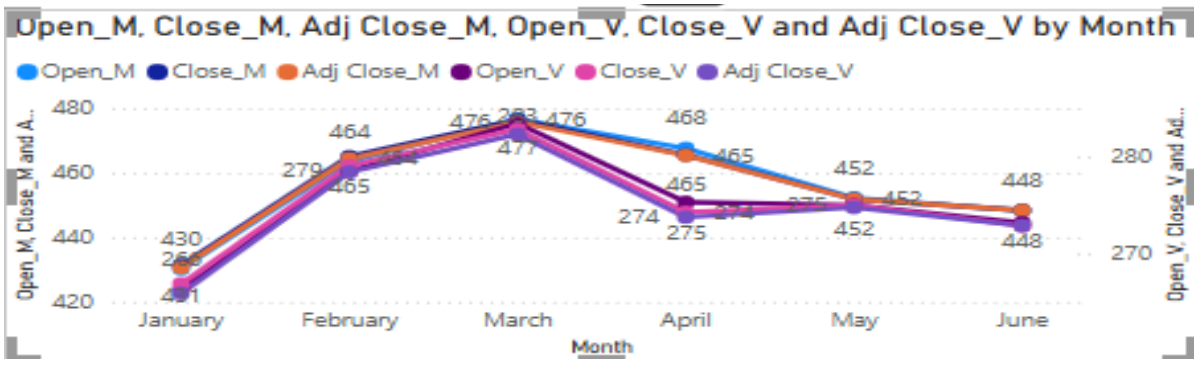


Chart 3.2. Noticeable drop

- According to Chris Suh, the Chief Financial Officer of Visa, "In the U.S., while growth in the high-spend consumer segment remained stable compared to prior quarters, we saw a slight moderation in the lower-spend consumer segment.".
- Moreover, Visa was also affected by rival companies like American Express (AXP.N) missing revenue expectations in quarter 2. Michael Ashley Schulman, the Chief Investment Officer at Running Point Capital Advisors, remarked, "Visa was priced for perfection back in March but has eased since then as unemployment, payment and loan delinquencies, and continued consumer disposable income concerns tick higher." (Arasu, 2024)
- Since the data only goes up to 2024, no conclusions can be drawn about the stock trends for 2024.
- This is the reason why the line chart experienced the noticeable decrease in quarter 2.

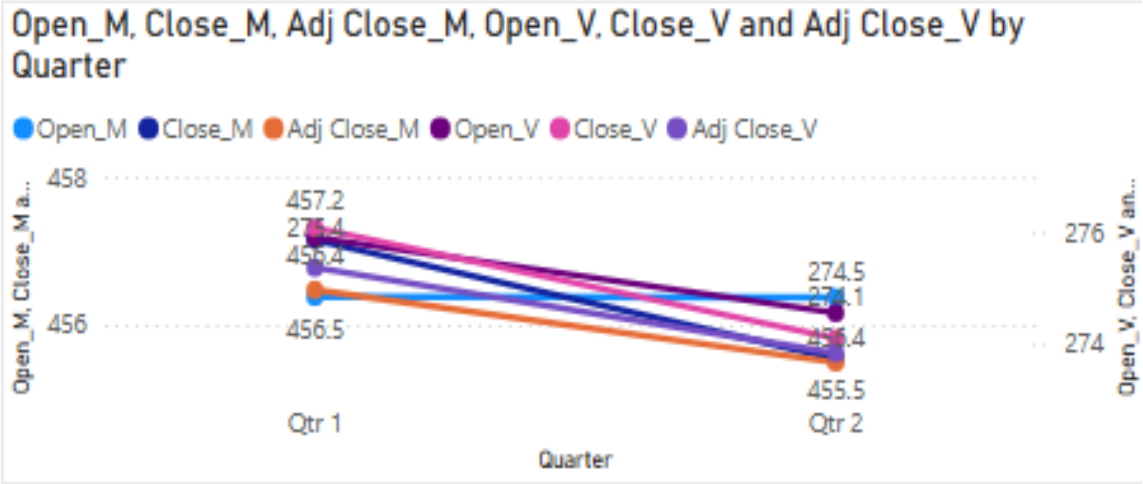


Chart 3.3. Noticeable decrease in quarter 2

- MasterCard Stock Trends: Similar to Visa, MasterCard's stock price has shown significant growth, especially after the 2010s, with large fluctuations during certain periods. The forecast for MasterCard stock indicates a sharp growth trajectory until around 2024. Similar to Visa, since the data only goes up to 2024, no conclusions can be drawn about the stock trends for 2024.

II. Trading Volume Trends for Visa and MasterCard

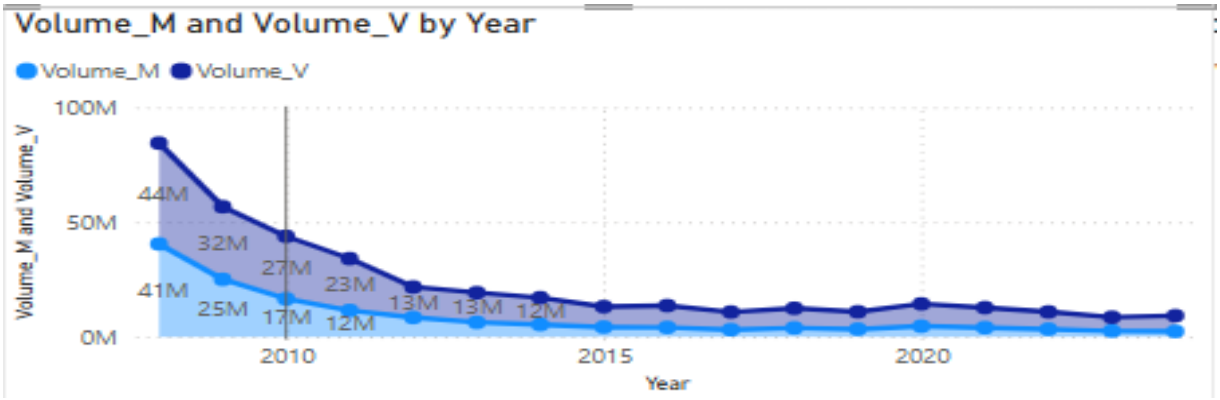


Chart 3.4. Trading Volume Trends for Visa and MasterCard

- Visa Trading Volume: The trading volume for Visa has shown a steady decrease from 2008 onward, with notable periods of high activity around the 2010s, followed by a general downtrend.
- MasterCard Trading Volume: Similarly, MasterCard's trading volume has followed a decreasing trend since 2008. There were some spikes in trading activity in the early 2010s, but the volume has decreased progressively.
- Additionally, it can be observed that from 2008 to 2010, Visa’s stock trading volume was significantly higher than that of MasterCard, but gradually, the volumes of these two stocks have become almost equal.
- One possible reason for this decrease is that Visa’s stock was listed on the stock exchange later than MasterCard’s, so initially, there were more investors attracted to Visa's stock.

III. High and Low Price Fluctuations for Visa and MasterCard

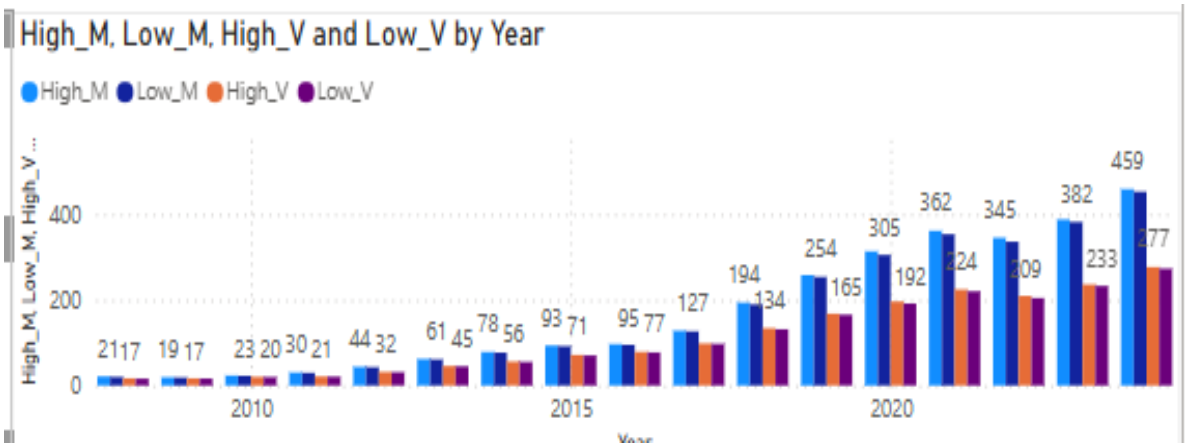


Chart 3.5. High and Low Price Fluctuations for Visa and MasterCard

- The highest and lowest price fluctuations for both Visa and MasterCard are impacted by broader economic trends. Both stocks exhibit substantial volatility, particularly around 2010-2015. However, these fluctuations become less pronounced after 2020, reflecting greater market stability for both companies.
- One reason for this fluctuation could be that starting in 2020, the COVID-19 pandemic caused significant losses to the global economy, leading financial markets to experience substantial volatility. However, after the situation was brought under control, the markets gradually stabilized, reducing the volatility of stocks like Visa and MasterCard, as these companies adapted and maintained a stronger position.
- Visa's price lows tend to correlate with economic downturns, while MasterCard's low points show greater sensitivity to financial crises or market corrections. This suggests that Visa's stock is more resilient during broader economic challenges, while MasterCard may be more vulnerable to financial market shocks.

IV. Relation Between Trading Volume and Price Movements

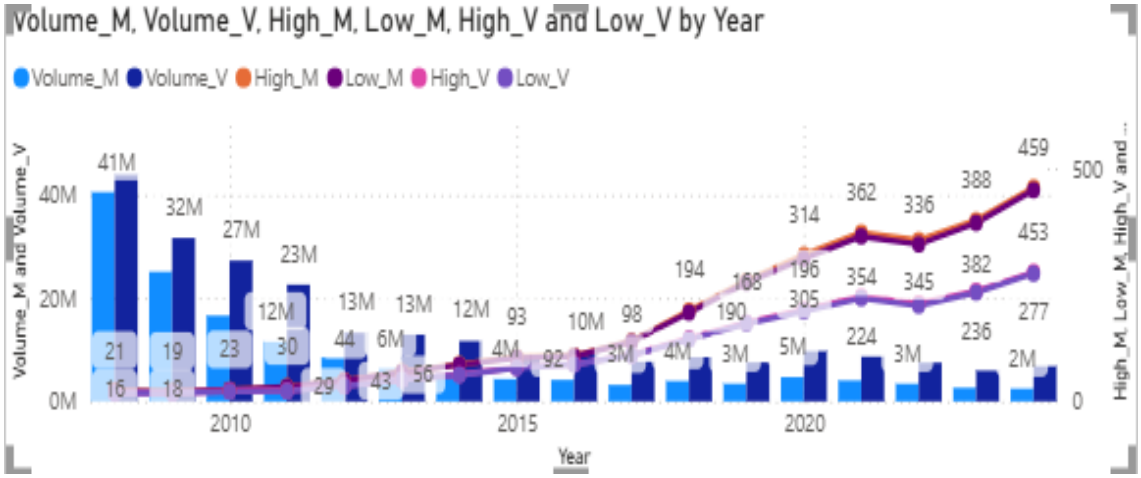


Chart 3.6. Relation Between Trading Volume and Price Movements

- Although the trading volume of MasterCard is lower than Visa's, the high price and low price of MasterCard are higher and exhibit more significant fluctuations than those of Visa.
- This could be because trading volume doesn't always have relation with price volatility. Factors like market sentiment, impact of news, or stock-specific characteristics can cause more dramatic price fluctuations, even if trading volume remains lower.
- The strong price volatility may reflect the market's sensitivity to information or events related to MasterCard, leading to larger swings in stock prices during critical periods.

V. Forecasted Stock Prices for Visa and MasterCard

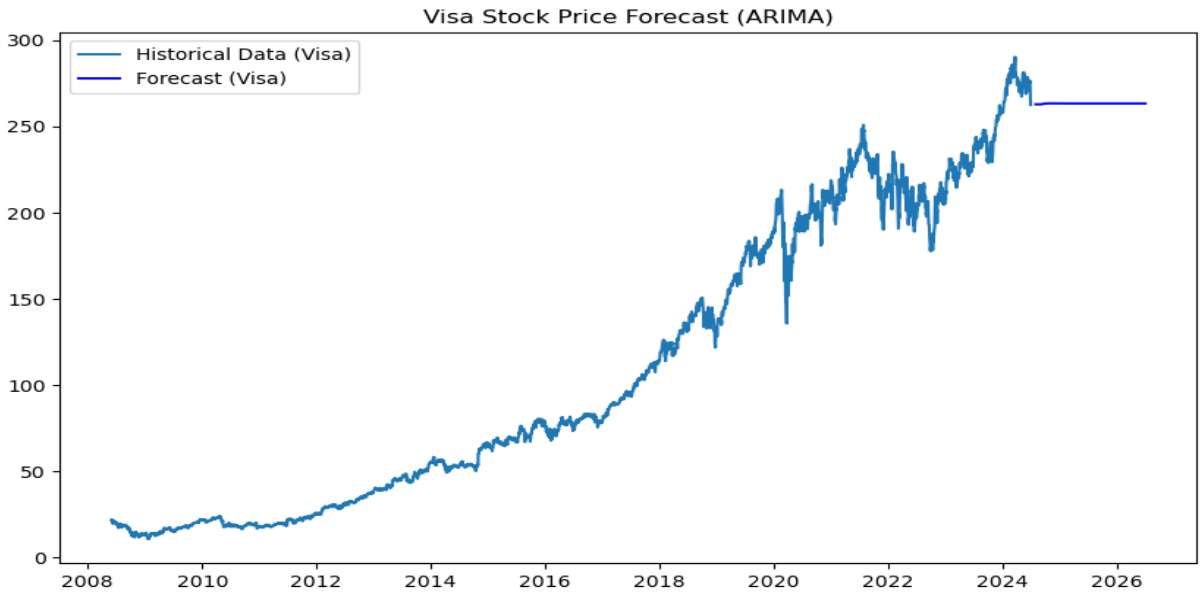


Chart 3.7. Forecasted Stock Prices (Close Price) for Visa

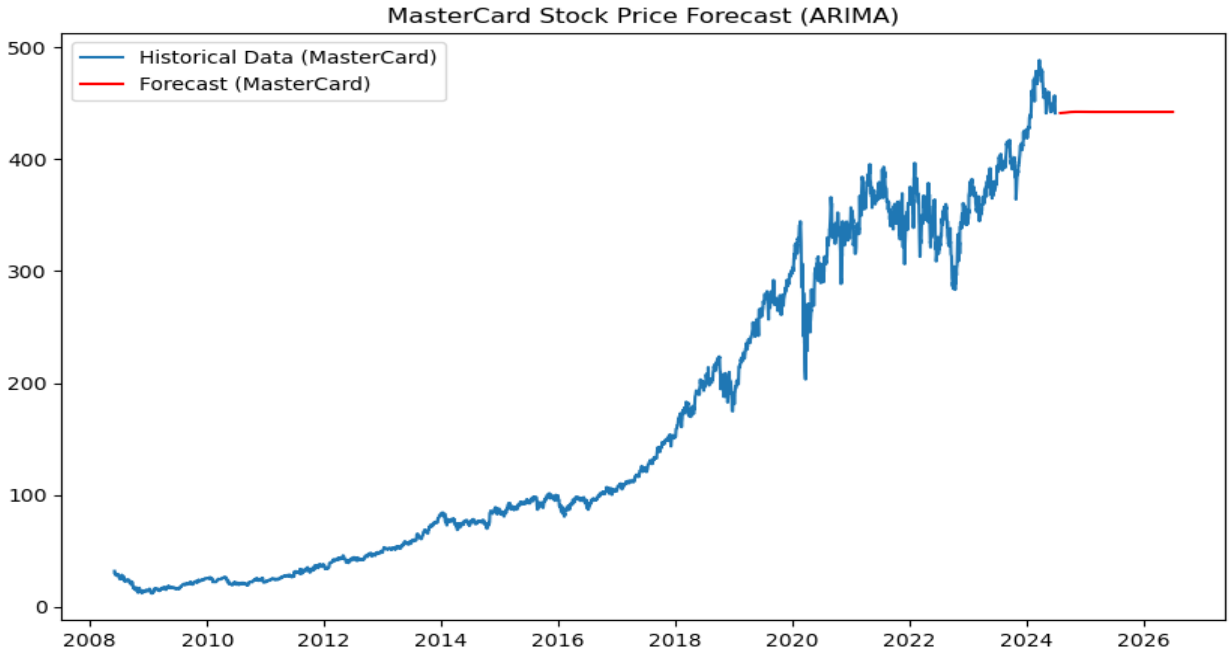


Chart 3.8. Forecasted Stock Prices (Close Price) for MasterCard

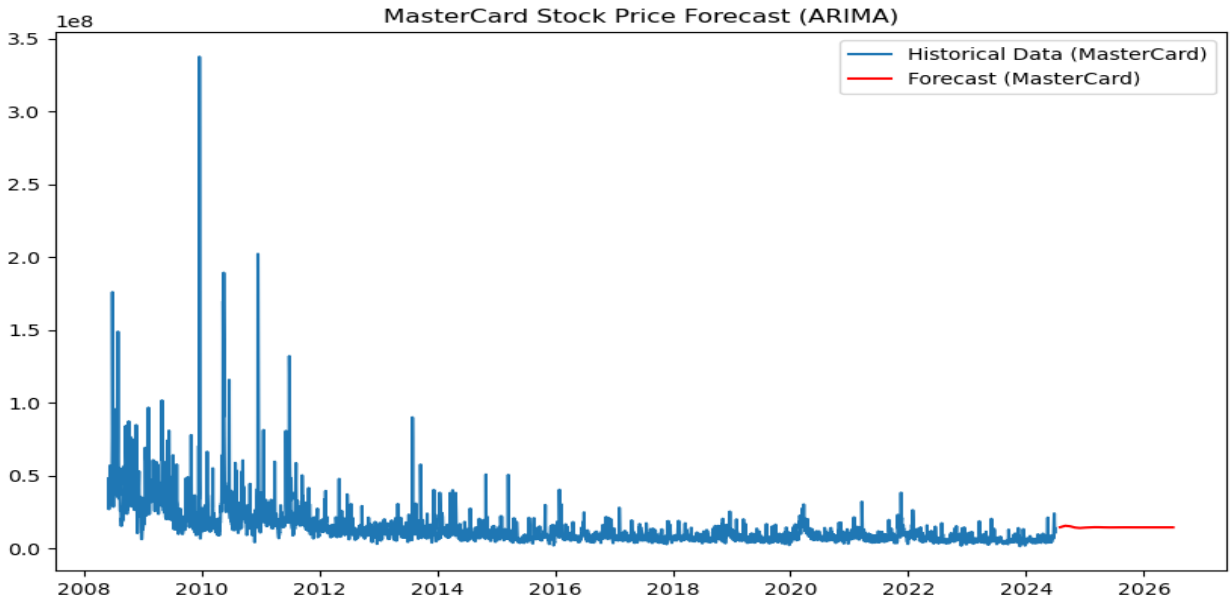


Chart 3.9. Forecasted Stock Prices (Volume) for MasterCard

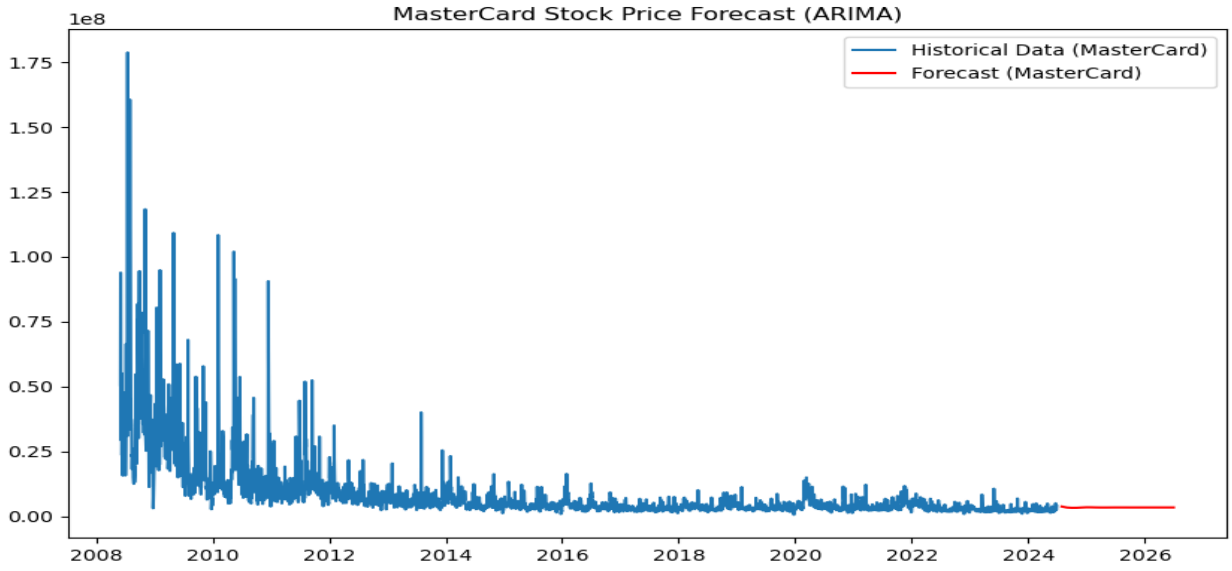


Chart 3.10. Forecasted Stock Prices (Volume) for MasterCard

- The stock price forecast indicates that both Visa and MasterCard are expected to remain stable, with no significant changes in price after June 2024.
- Both companies are projected to maintain stability in the short term, with no major fluctuations.

- This suggests that investors can expect the stocks of Visa and MasterCard to remain steady in the near future, although there won't be strong growth.
- However, attention should still be given to external factors that could impact the market in the future, including global economic factors or changes in consumer behavior.

VI. Recommended Investment Actions (Buy, Hold, Sell)

- Visa: Based on historical data and forecast trends, Hold seems to be the most appropriate action for Visa. The stock has shown consistent growth, but the forecast predicts limited gains in the near future. Investors might consider holding Visa stocks for stability rather than expecting significant growth in the coming years.
- MasterCard: For MasterCard, the recommendation is also to Hold based on the historical data showing stable growth over the years. However, the forecast indicates potential instability, particularly in 2024, so it might be prudent for investors to wait and assess market conditions before making drastic moves.

VII. Risks and Opportunities in the Forecast Data

- Opportunities: Both Visa and MasterCard present opportunities for investors who seek stability. However, given the predicted flattening of stock price growth, investors should be cautious of investing heavily at the peak of the current market trend. Visa's ongoing global expansion presents a strong case for long-term investment, though the recent flat forecast could signal market maturity.
- Risks: The biggest risk arises from the forecast for MasterCard, where a significant dip in stock price is predicted. This suggests that investors should be wary of buying into MasterCard at this stage, as external factors could trigger a market correction.

VIII. Parts of the assignment used AI and the purpose of its use

- For this project, I mainly used AI in Chapter 3.
- I used AI in combination with other online sources to compare stock price fluctuations.
- Initially, I referred to various sources and noticed that the Close price is the most important factor. Some sources used only the Close price, while others used both the Open and Close prices.

- Then, I consulted ChatGPT Plus and Gemini and found that it is possible to use Open, Close, and Adj Close (which, as I understand, refers to the adjusted closing price of stock after accounting for factors such as dividends, etc.).
- Additionally, I used AI to write Python code to run prediction models and decision-making models.
- Initially, I planned to use Power BI to create forecasts, but due to either my Power BI version being outdated (I updated to the new version in May 2025) or other reasons, despite selecting Analytics in the line chart, the forecast option did not appear.
- Furthermore, when writing the findings, I also referred to GPT and sources such as Reuters and Yahoo Finance to provide explanations for those findings.

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Arasu, B. K. (2024). *Visa reports rare quarterly revenue miss, shares drop.* Retrieved from Reuters: https://www.reuters.com/business/finance/visa-quarterly-profit-jumps-20-strong-consumer-spending-2024-07-23/